# SECOND-ORDER RESPONSE: DC MOTOR ANGLE UNDER P CONTROL

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## 1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

**Table 1:** *Learning Outcomes for the Practical Exercise*

| ID | Learning Outcome | Achieved |
|---|---|---|
| 1 | Derive the closed-loop transfer function of proportional angle control of the DC motor, and express its damping ratio and natural frequency in terms of the gain. | ☐ |
| 2 | Choose proportional gains for target damping ratios, and measure the overshoot, peak time and settling time of the resulting step responses. | ☐ |
| 3 | Back-calculate the damping ratio and natural frequency from measured responses, and explain any differences from the prediction. | ☐ |

## 2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The angle of the shaft is measured from the encoder counts, as described in the practical exercise, Manual PID Tuning of DC Motor Angle Control, and the output voltage of the controller sets both the direction and the PWM value. Because the angle is the integral of the speed, the angle model is $G_\theta(s) = K_\theta/(s(\tau s + 1))$, where $K_\theta = 6K$ converts RPM/V into degrees per second per volt. For the A1.4 example motor, $K_\theta = 168$ °/(s·V).

## 3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A proportional controller on the shaft angle turns the first-order motor into a second-order closed loop. Its damping ratio and natural frequency are predicted from the

motor model and then measured from step responses. The exercise supports the ACE-CORE learning guide, Response Analysis of Second-Order Transfer Functions.

### Derive the Second-Order Closed Loop

With the angle model $G_\theta(s) = K_\theta/(s(\tau s + 1))$ and proportional control with unity feedback, the closed-loop transfer function is:

$$T(s) = \frac{K_p K_\theta}{\tau s^2 + s + K_p K_\theta} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n = \sqrt{\frac{K_p K_\theta}{\tau}}, \qquad \zeta = \frac{1}{2\sqrt{K_p K_\theta\, \tau}}$$

For an underdamped response ($\zeta < 1$), the standard measures follow from $\zeta$ and $\omega_n$:

$$OS = 100\, e^{-\pi\zeta/\sqrt{1-\zeta^2}}\ \%, \qquad t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}, \qquad t_s \approx \frac{4}{\zeta\omega_n}$$

Increasing $K_p$ raises $\omega_n$ but lowers $\zeta$: the response becomes faster but more oscillatory. Rearranging the expression for $\zeta$ gives the gain for a target damping ratio, $K_p = 1/(4\zeta^2 K_\theta \tau)$. For a critically damped response ($\zeta = 1$) the settling time is approximately $5.83/\omega_n$ instead.

The gains for three target damping ratios, and the predicted responses, are given in Table 2 for the example motor.

**Table 2:** *Predicted Responses for the A1.4 Example Motor ($K_\theta$ = 168 °/(s·V), $\tau$ = 0.15 s)*

| Target $\zeta$ | $K_p$ (V/°) | $\omega_n$ (rad/s) | OS (%) | $t_p$ (s) | $t_s$ (s) |
|---|---|---|---|---|---|
| 1.0 | 0.0099 | 3.33 | 0.0 | – | 1.75 |
| 0.5 | 0.0397 | 6.67 | 16.3 | 0.54 | 1.20 |
| 0.3 | 0.1102 | 11.11 | 37.2 | 0.30 | 1.20 |

### Measure the Step Responses

**[Screenshot to insert]** *Step (angle setpoint, deg) → Sum (+ −) → Gain Kp → Saturation ±5 V → sign to direction pin, |u|·255/5 to PWM; encoder count → angle (360·n/CPR) → feedback. Scopes log r, θ, u and t.*

**Figure 1:** *Simulink Model for Proportional Angle Control*

1. Build the model in Figure 1 with $T_s$ = 0.01 s. Recalculate the gains in Table 2 with your own $K_\theta$ and $\tau$.
2. Apply a 30° step at $t$ = 0.5 s for each gain and log the setpoint, angle, output voltage and time. A 30° step keeps the output below the 5 V limit at the highest gain.

3. Measure the overshoot, peak time and settling time of each response, and back-calculate $\zeta$ from the overshoot and $\omega_n$ from the peak time with the script below.
4. Record the results in Table 3.

**MATLAB Script:** back-calculating the damping ratio and natural frequency

```
th = theta.signals.values(:);  t_k = t.signals.values(:);
t0 = 0.5;  R = 30;                                % step time [s] and size [deg]
idx = t_k >= t0;  tn = t_k(idx) - t0;  yn = th(idx);
y_inf = mean(yn(tn > tn(end) - 0.5));
[y_max, i_max] = max(yn);
OS = (y_max - y_inf)/y_inf*100;                   % overshoot [%]
tp = tn(i_max);                                   % peak time [s]
L = log(OS/100);
zeta = -L/sqrt(pi^2 + L^2)                        % from the overshoot
wn = pi/(tp*sqrt(1 - zeta^2))                     % from the peak time [rad/s]
```

Example responses are shown in Figure 2. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares ($K$ = 28 RPM/V, $\tau$ = 0.15 s), so they show the behaviour to expect rather than measured data, integrated to give the angle. The simulated overshoot, about 21% and 50% for $\zeta$ = 0.5 and 0.3, is higher than the continuous-time prediction, because the digital controller samples every 0.01 s and its output takes effect one sample later, which adds lag to the loop. On the motor, friction acts the other way and adds damping, so your measured overshoot may fall either side of the prediction, and friction can also leave a small final error.

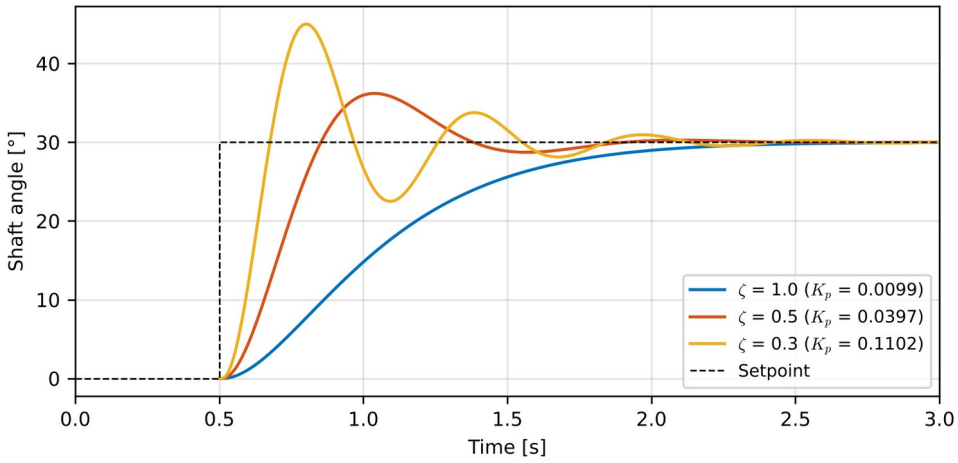



**Figure 2:** *Simulated Example of the Angle Response to a 30° Step for Three Target Damping Ratios*

**Table 3:** *Measured Responses for Your Motor*

| Target $\zeta$ | $K_p$ (V/°) | OS (%) | $t_p$ (s) | $t_s$ (s) | Measured $\zeta$ | Measured $\omega_n$ (rad/s) |
|---|---|---|---|---|---|---|
| 1.0 | | | – | | – | – |
| 0.5 | | | | | | |

| Target $\zeta$ | $K_p$ (V/°) | OS (%) | $t_p$ (s) | $t_s$ (s) | Measured $\zeta$ | Measured $\omega_n$ (rad/s) |
|---|---|---|---|---|---|---|
| 0.3 | | | | | | |

1. Why does the angle loop reach the setpoint with proportional control alone, when the speed loop in the practical exercise, Final Value Theorem and Steady-State Error, did not?
2. Which single physical change to the motor would raise $\zeta$ for the same $K_p$?
3. How would the response change if the output reached the 5 V limit during the step?

## 4.0 Concluding Remarks

This practical exercise showed that proportional control of the motor angle produces a second-order closed loop, and related its damping ratio and natural frequency to the controller gain. Using MATLAB, Simulink and ACE-Lab hardware, gains were chosen for three target damping ratios and the overshoot, peak time and settling time were measured.

The measured responses followed the second-order predictions: higher gain gave a faster, more oscillatory response. Back-calculating the damping ratio and natural frequency from the measurements, and explaining the differences through effects such as friction, links the textbook second-order system to the behaviour of real hardware.

## Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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